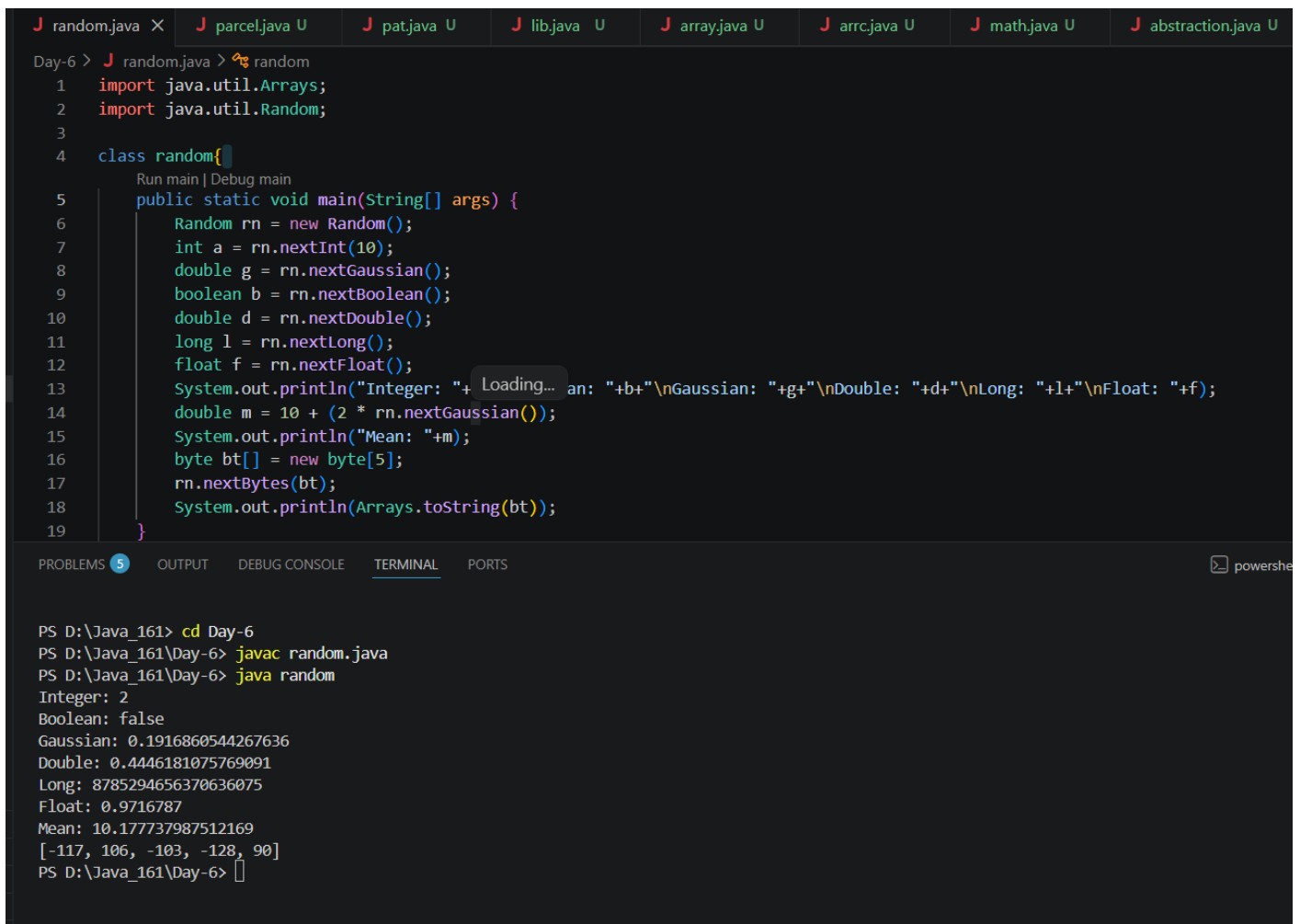


Task 1:

Write a Java program using Random to generate:

1. **Student ID** → `nextInt(1000)` (0–999)
2. **Marks** → `nextDouble()` * 100
3. **Attendance** → `nextFloat()` * 100
4. **Status** → `nextBoolean()` (Pass/Fail)
5. **Registration Number** → `nextLong()`
6. **Performance Score** → `nextGaussian()` * 10
7. **Random Bytes** → `nextBytes()` (array size 5)
8. **5 Random Numbers** → `ints()` stream



```
random.java X parcel.java U pat.java U lib.java U array.java U arrc.java U math.java U abstraction.java U
Day-6 > random.java > random
1 import java.util.Arrays;
2 import java.util.Random;
3
4 class random{
5     public static void main(String[] args) {
6         Random rn = new Random();
7         int a = rn.nextInt(10);
8         double g = rn.nextGaussian();
9         boolean b = rn.nextBoolean();
10        double d = rn.nextDouble();
11        long l = rn.nextLong();
12        float f = rn.nextFloat();
13        System.out.println("Integer: "+ a + "\nGaussian: "+g+"\nDouble: "+d+"\nLong: "+l+"\nFloat: "+f);
14        double m = 10 + (2 * rn.nextGaussian());
15        System.out.println("Mean: "+m);
16        byte bt[] = new byte[5];
17        rn.nextBytes(bt);
18        System.out.println(Arrays.toString(bt));
19    }
}
```

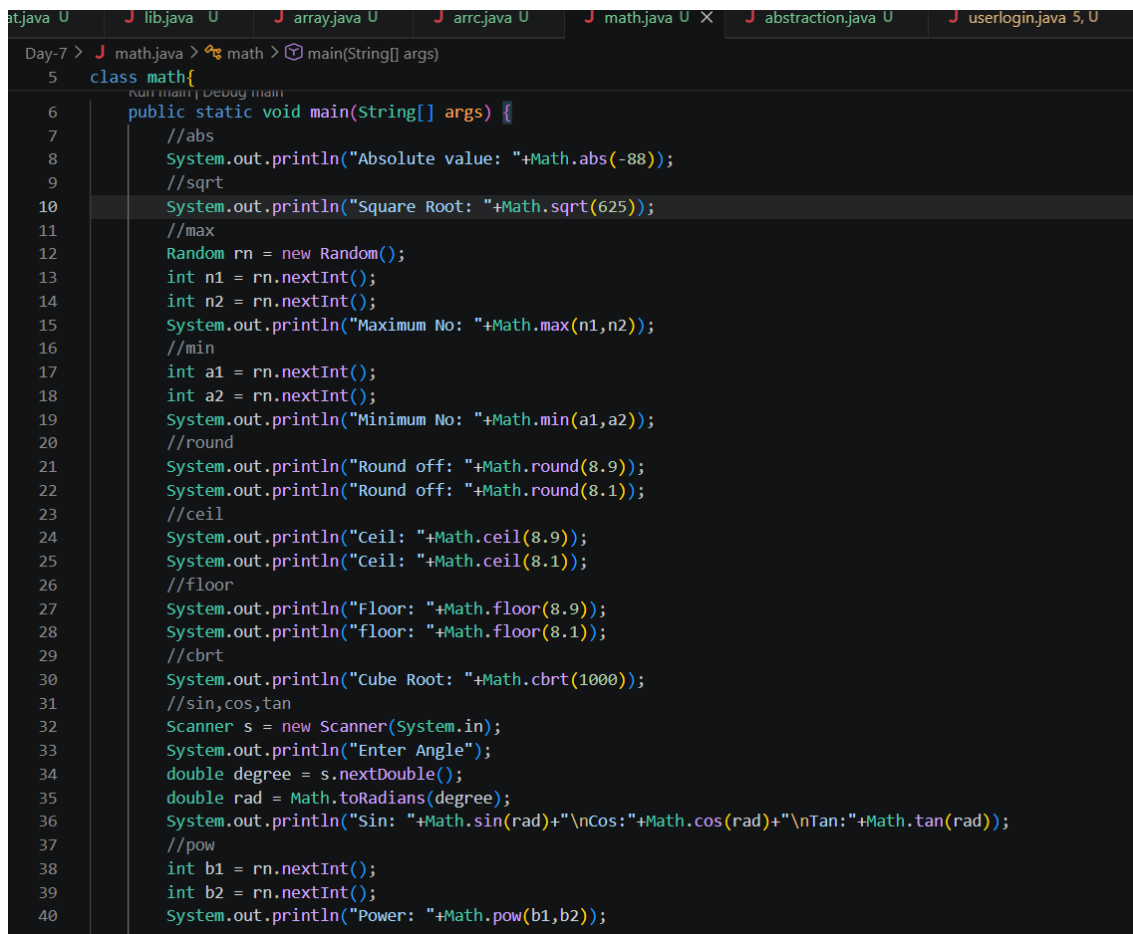
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS D:\Java_161> cd Day-6
PS D:\Java_161\Day-6> javac random.java
PS D:\Java_161\Day-6> java random
Integer: 2
Boolean: false
Gaussian: 0.1916860544267636
Double: 0.4446181075769091
Long: 8785294656370636075
Float: 0.9716787
Mean: 10.177737987512169
[-117, 106, -103, -128, 90]
PS D:\Java_161\Day-6>
```

Task 2:

Write a Java program using Math class to perform the following:

1. Find Maximum & Minimum marks → Math.max(), Math.min()
2. Round the marks → Math.round()
3. Find Square & Square Root → Math.pow(), Math.sqrt()
4. Find Absolute difference → Math.abs()
5. Find Ceiling & Floor values → Math.ceil(), Math.floor()
6. Find random number (0–100) → Math.random()
7. Find exponential value → Math.exp()
8. Find logarithm → Math.log()
9. Find sine & cosine values → Math.sin(), Math.cos()
10. Find cube of a number → Math.pow(num, 3)



```
Day-7 > J math.java > math > main(String[] args)
5 class math{
6     public static void main(String[] args) {
7         //abs
8         System.out.println("Absolute value: "+Math.abs(-88));
9         //sqrt
10        System.out.println("Square Root: "+Math.sqrt(625));
11        //max
12        Random rn = new Random();
13        int n1 = rn.nextInt();
14        int n2 = rn.nextInt();
15        System.out.println("Maximum No: "+Math.max(n1,n2));
16        //min
17        int a1 = rn.nextInt();
18        int a2 = rn.nextInt();
19        System.out.println("Minimum No: "+Math.min(a1,a2));
20        //round
21        System.out.println("Round off: "+Math.round(8.9));
22        System.out.println("Round off: "+Math.round(8.1));
23        //ceil
24        System.out.println("Ceil: "+Math.ceil(8.9));
25        System.out.println("Ceil: "+Math.ceil(8.1));
26        //floor
27        System.out.println("Floor: "+Math.floor(8.9));
28        System.out.println("floor: "+Math.floor(8.1));
29        //cbt
30        System.out.println("Cube Root: "+Math.cbrt(1000));
31        //sin,cos,tan
32        Scanner s = new Scanner(System.in);
33        System.out.println("Enter Angle");
34        double degree = s.nextDouble();
35        double rad = Math.toRadians(degree);
36        System.out.println("Sin: "+Math.sin(rad)+"\nCos: "+Math.cos(rad)+"\nTan: "+Math.tan(rad));
37        //pow
38        int b1 = rn.nextInt();
39        int b2 = rn.nextInt();
40        System.out.println("Power: "+Math.pow(b1,b2));
41    }
```

```
PS D:\Java_161\Day-7> javac math.java
PS D:\Java_161\Day-7> java math
Absolute value: 88
Square Root: 25.0
Maximum No: -77251836
Minimum No: -1805449639
Round off: 9
Round off: 8
Ceil: 9.0
Ceil: 9.0
Floor: 8.0
floor: 8.0
Cube Root: 10.0
Enter Angle
1
Sin: 0.01745240643728351
Cos:0.9998476951563913
Tan:0.017455064928217585
Power: Infinity
PS D:\Java_161\Day-7> █
```

Task:3

Scenario:

A bank system allows users to perform basic account operations like deposit and withdrawal using abstraction.

Task:

- Initial Balance = 0

Create an **abstract class** Account with:

- Abstract methods:
 - deposit(double amount)
 - withdraw(double amount)

Create a subclass MyAccount to:

1. Add amount to balance using deposit()

2. Subtract amount from balance using withdraw()
3. Maintain and update balance properly

Operations to Perform:

1. Deposit = 1000
2. Withdraw = 400

Expected Output:

Deposited: 1000

Withdrawn: 400

Balance: 600

```
1  import java.util.Scanner;
2
3
4  class abstraction{
5      Run main | Debug main
6      public static void main(String[] args) {
7          int pin_no = 1512;
8          Scanner s = new Scanner(System.in);
9          bank_server bs = new bank_server();
10         System.out.print("Enter your pin: ");
11         int pin = s.nextInt();
12         int choice;
13         if(pin_no != pin){
14             System.out.println("Wrong Pin");
15             System.exit(0);
16         }
17         do{
18             System.out.println("Enter your choice: \n1.Check Balance \n2.Deposite \n3.withdrawl: \n4.Enter Zero to exit");
19             System.out.print("Your Choice: ");
20             choice = s.nextInt();
21             if(choice == 1){
22                 System.out.println("Balance amt: "+bs.getbalance());
23             }
24             else if(choice == 2){
25                 System.out.print("Enter your amt: ");
26                 double a = s.nextDouble();
27                 bs.deposite(a);
28             }
29             else if(choice == 3){
30                 System.out.print("Enter your amt: ");
31                 double b = s.nextDouble();
32                 bs.withdrawl(b);
33             }
34             else{
35                 System.out.println("Process Completed..");
36             }
37         }
38     }
39 }
```

Day-7 > abstraction.java > bank_server

```
4  class abstraction{
5      public static void main(String[] args) {
34         else{
35             System.out.println("Process Completed..");
36         }
37     }while(choice!=0);
38
39     }
40
41 }
42
43 class bank_server{
44     private double balance = 5000.7;
45     double getbalance(){
46         return balance;
47     }
48     void deposit(double amt){
49         balance += amt;
50     }
51     void withdrawl(double amt){
52         if(balance<=amt){
53             System.out.println("Insufficient balance");
54         }
55         else
56             balance -= amt;
```

PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Power: Infinity

PS D:\Java_161\Day-7> javac abstraction.java

PS D:\Java_161\Day-7> java abstraction

Enter your pin: 1512

Enter your choice:

1.Check Balance

2.Deposite

3.withdrawl:

4.Enter Zero to exit

Your Choice: 1

Balance amt: 5000.7